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Online Distributed Routing Problem of Electric Vehicles

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***Abstract*— Considering the penetration of numerous electric vehicles (EV) into the transportation sector, the EV routing problem that jointly optimizes the charging and routing process of EVs is becoming increasingly popular, which should be solved in online fashion for the practical requirements. First, we start with an offline EVs routing and charging (EVRP) problem which is a large-scale mixed integer nonlinear program (MINLP). Considering the NP hardness of MINLP, solving the offline EVRP directly is a time-consuming task that is not suitable for the online setting. Hence, a Benders decomposition based method is proposed to decompose the offline EVRP into a master problem and a set of sub-problems which allows for distributed implementation. To further speed up the computation, we relax the mixed integer master problem to one that can be equivalently solved as a linear program. Moreover, a novel kind of valid cut is added to the relaxed master problem to further reduce the number of iterations. In order to adapt the offline EVRP to the online setting, by introducing the virtual depot, we utilize the rolling-horizon framework to tackle the uncertainty of future information, where the offline EVRP is solved in real-time repeatedly. Finally, simulations using real road map in Belgium are performed. Besides, numerical results validate that the computation speed of proposed algorithm is faster than the state-of-the-art algorithms by several orders of magnitude, and showcase the capability of proposed algorithm to solve large size instances up to 350 nodes with 35 EVs within 100 seconds.**

***Index Terms*— Electric vehicle routing problem, charging problem, distributed optimization, Benders decomposition, online algorithm.**

NOMENCLATURE

Given Parameters

$\mathcal{K}$	The set of EVs.
$\mathcal{R}$	The set of transportation request nodes.
$\mathscr{M}_i$	Revenue from serving requests when $i \in \mathcal{R}$, set as negative value.
ω_T	The discount factor for the total charging time.
ω_T	The discount factor for the total travel time.
c_{ev}	EVs' usage cost.
d_{ij}	Travel distance from node i to j where $i, j \in \mathcal{V}$.
e_{ij}	Energy consumption traveling from node i to node j, $e_{ij} = \epsilon d_{ij}$, where ϵ is the energy consumption rate.
g_i	Charging duration per unit power (1 kWh) at node $i, i \in \mathcal{V}$.
p_i	Charging price at node $i, i \in \mathcal{V}$.
t_i	Pickup time of transportation requests at node $i, i \in \mathcal{R}$ given by customers.
T_{ij}	Travel time from node i to j where $i, j \in \mathcal{V}$.
t_i^{ocr}	The occurrence time of request i.

Decision variables

r_i^k	Continuous decision variable indicating charged energy at node $i, i \in \mathcal{V}$.
x_{ij}	Binary decision variable indicating whether edge $(i, j) \in \mathcal{E}$ is selected as a part of the solution or not.
x_{ij}^k	Binary decision variable indicating whether $k \in \mathcal{K}$ traverses $(i, j) \in \mathcal{E}$ or not.

Other Symbols

$\tilde{E}_i^k$	The copied variable of E_i^k.
E_i^k	Battery energy level of $k, k \in \mathcal{K}$ at node $i, i \in \mathcal{V}$.
$G'(\mathcal{V}', \mathcal{E}')$	Directed graph, the node set $\mathcal{V}'$ represents geographical nodes.
$G(\mathcal{V}, \mathcal{E})$	Directed graph, $V = \{v_1, v_n\} \cup \mathcal{R}$ includes the start depot. end depot and transportation request nodes.

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I. INTRODUCTION

NOWADAYS, the transportation industry is going through a major evolution. In order to tackle the energy shortage and pollution issues, many countries are developing an environmental-friendly transportation system consisted of lots of electric vehicles (EVs). According to [1], compared with 45 thousand EVs sold in 2011,[1] sales of electric cars topped 2.1 million globally in 2019. To handle the penetration of massive EVs, some issues naturally emerge, e.g., uncoordinated charging phenomenon, and road congestion. Thus, designing

[1]Source: https://www.ourenergypolicy.org/wp-content/uploads/2013/09/GlobalEVOutlook_2013.pdf

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TABLE I
COMPARISON BETWEEN RELATED WORKS

References	Model	Objective function	Solution Approach
Kancharla et al. [26]	MIP	Travel time, charging time	The Adaptive Large Neighborhood Search (ALNS) algorithm
Zhu et al. [27]	MIP	Travel time, service time, charging time, and waiting time	An improved neighbor routing initialization method, the adaptive elitist genetic algorithm
Lee [28]	MIP	Travel distance, and charging time	Branch-and-price approach
Liang et al. [29]	MIP	Total duration (cost)	Branch-and-price-and-cut method, tabu search
Yang et al. [30]	MIP	Travel distance	The augmented Lagrangian relaxation method, and the time-dependent forward dynamic programming algorithm.
Our paper	MINLP	Traveling time, charging time, charging price, EVs usage cost, and revenue from serving customers	Modified Benders decomposition method, and Lagrangian relaxation method

a decent routing and charging schedule of multiple EVs is beneficial for the EVs' operator and in some senses is also a challenging task, which is required to be solved in an efficient manner for real time implementation.

As a classic combinatorial optimization problem, vehicle routing problem (VRP) has been well researched [2]. Furthermore, the limited battery capacity and considerable charging time of EVs render the VRP problem even more challenging, thus leading to a new research area named electric vehicle routing problem (EVRP) [3] that jointly optimizes the charging and routing process of EVs. In practice, transportation requests and future traffic information are usually uncertain, and typically reveal dynamically over time. Clearly, tackling the EVRP in an online setting is more valuable and promising in transportation applications. Therefore, in this work, we focus on the online EVRP problem.

A. Related Work

1) Electric Vehicles Pouting Problem: In recent years, there have been extensive research efforts put on EVRP [4]–[23]. For instance, the authors of [20] focus on the time-dependent EVRP with chance-constraints and partial recharging, in which the routing method is divided into two stages, *1)* finding the best paths and *2)* optimizing the routes. Zang *et al.* [21] propose a detailed mathematical model regarding electric vehicles routing problem including the electricity cost, the total battery depreciation cost, and the drivers' salaries. By exploiting the power of the learning approaches, an end-to-end learning based algorithm proposed in [22] is aimed for solving the EVRP with time windows. By adopting the new electricity replenishment technology, battery swapping technology, Raeesi and Zografos [24] attempt to solve the EVRP with time windows and synchronised mobile battery swapping. In addition, to solve the capacitated EVRP (CEVRP), Jia *et al.* [25] propose to divide the CEVRP into two levels of subproblem: 1) capacitated VRP and 2) fixed route vehicle charging problem. For better illustration, the characteristics of few works are presented in Table I.

The aforementioned papers [20]–[22], [24]–[30] all neglect the energy or electricity cost in their objective function, which is a crucial part to consider in the routing and charging problem of EVs. With one exception [21], Zang *et al.* assume that the electricity cost is linearly dependent on the travel distance, which is unpractical in practice. In this paper, contrary to these papers, we explicitly model the energy or electricity cost as the product of charging energy and charging price.

2) Exact Approaches: In order to guarantee the solution quality, researchers propose to solve the EVRP with mathematical programming techniques [5], [6], [14]–[16], [31] e.g., decomposition method, column generation method, mixed integer programming (MIP) based approach. Specifically, [16] reformulates the single EV routing problem as the minimum cost flow problem solved by simplex method while there still is an MIP problem in the case of multiple EVs. In order to provide an optimal routing plan for the shuttle fleet that serves to a university settlement, Hulagu and Celikoglu [32] use mixed integer program (MIP) model to capture the features of aforementioned problem. However, to obtain an exact solution, there is an unbearable computation load in [32]. Recently, Benders decomposition method has been widely used in routing problem [33]–[37]. For instance, the authors in [35] propose an exact Benders decomposition approach to obtain the solution of production routing problem arising in the applications of integrated supply chain. However, the optimality gained from Benders decomposition results in the compromise of scalability of proposed approach, and the size of solvable instances in [35] is limited to 40 customers and 3 vehicles. Although the optimal solution can be obtained, these mathematical programming based approaches are not suitable for the online setting due to long computation time and unbearable computation load.

3) Approximation Methods: The approximation methods such as Hybrid Variable Neighborhood Search (HVNS) method [7], [38], Adaptive Large Neighborhood Search (ALNS) algorithm [8], [10], [39], Granular Tabu Search method (GTS) [9], are widely used in solving the EVRP [7]–[12]. There are also some papers [13], [17]–[19] working on convex relaxation heuristic method which converts the discrete optimization problem into a continuous one. In [19], considering diesel-driven vehicles rather than EVs, a modified mixed integer optimization formulation is proposed for the taxi routing problem. To solve the routing and charging scheduling problem for internet of EVs, [13] proposes a distributed algorithm by relaxing the coupling constraints. As for the time-dependent electric vehicle routing problem, due to the limited sizes of solvable instances (no more than 15 customers) solved by commercial solver (CPLEX), Lu *et al.* devise an iterated variable neighborhood search (IVNS) algorithm to find

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near-optimal solutions for larger instances [40]. In general, heuristic methods can obtain feasible solution with better performance on computation speed but without any guarantee on optimality.

Driven by the development of ride-hailing service and e-commerce, there are increasing interests of taxi company and truck transportation company in online EVRP. However, online EVRP is challenging to optimize due to 1) stochastic variables brought by online scheduling; 2) coupling relation between charging process and routing process of massive EVs; 3) NP-hardness of the formulated mathematical model. Different from aforementioned two kinds of approaches, we pursue a novel framework that solves the online EVRP in polynomial time and achieves near-optimal solution. To apply it in the online setting, we adopt the rolling-horizon approach, which repeatedly solves an offline EVRP with known customer requests and current traffic information, and implements the first time step actions as time goes by. As the offline EVRP needs to be solved in real time, a modified distributed Benders decomposition (MBD) algorithm is proposed to reduce its solution time complexity.

Statement of Contributions: Main contributions of this paper are presented as follows:

1) To tackle with the uncertainty of future information in the online EVRP, a rolling horizon approach that solves a corresponding offline EVRP repeatedly is proposed.
2) As the coupling between routing and charging problem renders the offline EVRP difficult to solve, we propose a modified Benders decomposition (MBD) framework to decompose the offline EVRP into several simpler problem, including a master problem and a set of subproblems that are linear programs and can be solved distributedly. After this decomposition, routing and charging are also decoupled.
3) Since the decomposed master problem is still a complex MILP, we devise a Lagrangian relaxation based method to relax a selected set of constraints, such that the remaining constraints happen to satisfy the total unimodularity property. With this technique, the relaxed master problem can be equivalently solved as a linear program. Despite that the original master problem is only solved approximately, we numerical validates its optimality gap which serves as a performance guarantee.
4) Moreover, a novel valid cut is devised to enhance the convergence of MBD. Acceleration effect of valid cut implies that proposed MBD achieves good performance in computation time reduction.
5) Virtual depot and an equivalent transformation are introduced. Only in this way, our proposed offline algorithm can be adapted to the rolling horizon framework and finally solve the online EVRP.

Organization: The paper is organized as follows. We elaborate the system models and mathematical formulation of online EVRP in Section II. To deal with the online EVRP, rolling horizon method is employed in which a corresponding offline EVRP is solved in real time repeatedly. Then BD and MBD are devised to solve the offline EVRP in Section III and IV respectively. The proposed algorithm is extended to the online setting in Section V. Section VI presents extensive simulation results to validate the effectiveness of proposed algorithm. Finally, section VII concludes this paper.

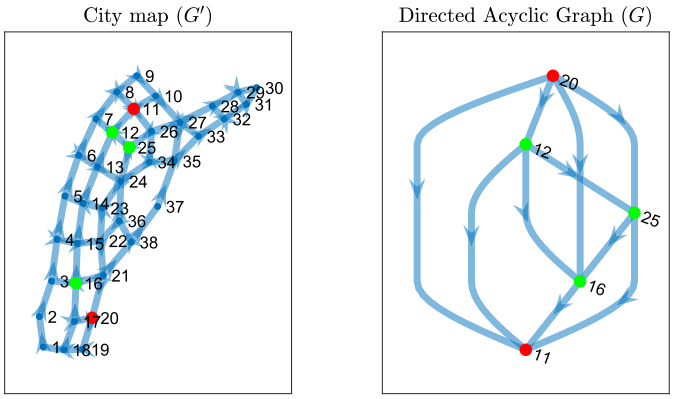


Fig. 1. An example of real road network and considered network. Red node 20 and 11 are selected as the start and end depot respectively, while the green node 12, 16, and 25 represent the transportation requests. The right directed acyclic graph is constructed by following the prescribed pickup or delivery time of transportation requests.

II. System Model and Problem Formulation

This section introduces the online EV routing problem. The system model is presented first, followed by the problem formulation.

A. System Model

Specifically, we consider a fleet of EVs traveling to serve massive customer transportation requests in a transportation network. Each customer request has a desired pickup time. By serving each request, an EV receives a certain amount of revenue. As in practice, the information of customer requests only reveals in real time, we shall consider the optimal online EVPR in this paper. Detailed system models are elaborated as follows:

1) Transportation Network: To reduce the number of nodes of road network, we adopt the preprocessing idea from [4] transforming $G'(\mathcal{V}', \mathcal{E}')$ which is a digraph consisted of geographical nodes, to a much smaller graph $G(\mathcal{V}, \mathcal{E})$. Note that $\mathcal{V} = \{v_1, \mathcal{R}, v_n\}$ represents the set of start depot v_1, transportation request nodes $\mathcal{R}$ as well as the end depot v_n, and $\mathcal{E}$ denotes the set of road segments with $(i, j) \in \mathcal{E}$ denoting a road link from vertex i to j. We denote d_{ij} and T_{ij} as the distance and travel time of road segments between vertices $i, j \in \mathcal{V}$ respectively.

2) Transportation Requests: A subset of vertices in graph G represents transportation requests node and is denoted as $\mathcal{R}$. In this paper, we use a tuple $\left\{t_i^{ocr}, t_i, \mathscr{M}_i\right\}, i \in \mathcal{R}$ to describe the characteristics of transportation requests. Here $\mathscr{M}_i$ is the revenue from serving request i. For each transportation request i, customer broadcasts their desired pickup time t_i at time t_i^{ocr}. For example, at 5:00, a customer broadcasts her transportation request that she needs to be picked up at 7:00, then t_i^{orc} is 5:00 while t_i is 7:00. In this paper, we assume that every transportation request vertex $i \in \mathcal{R}$ has charging facilities with charging duration per unit energy and price denoted by $g_i, p_i, i \in \mathcal{R}$ respectively. Note that if node i does not have

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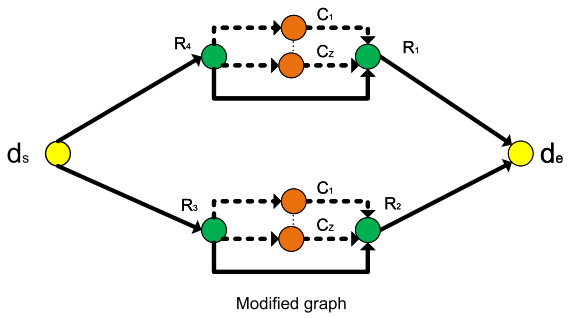


Fig. 2. The adoption of charging stations (CSs). $R_1, \cdots, R_4$ represent the request nodes, d_s, d_e denote the start and end depot, respectively, and $C_1, \cdots, C_Z$ represent the charging station nodes. The arcs connecting charging station nodes with request nodes are represented by the dashed arcs.

a charging facility, charging duration per unit energy g_i is simply set to ∞.

3) Electric Vehicles: The set of all EVs is denoted by $\mathcal{K}$. Each EV $k \in \mathcal{K}$ can be described by a tuple $\{E_0^k, E_{k,max}, c_{ev}\}$. E_0^k represents the initial energy of k starting from the depot, while the capacity of EV k is denoted as $E_{k,max}$. Meanwhile, there is also an EV usage fee c_{ev} needed to be considered due to monetary loss during the usage process of EV, such as wear and tear, depreciation, etc., which is quite common in related studies [14], [41]–[43]. All EVs station at the starting depot and return to the end depot. In this EVRP, each EV is assigned with a set of transportation requests to serve. With limited battery size, EV has to replenish at charging facility. The energy of EV k charged at node i is denoted by r_i^k. The maximum charging amount provided by charging facility is denoted by r_{max}.

Remark 1: The proposed model can also accommodate the case where the charging station and the transportation request are not the same node. We assume that there are Z charging stations in considered graph, which is illustrated in Fig. 2. Considering the original assumption that every request nodes are equipped with charging facilities, by following the subsequent procedures (i) setting the charging duration per unit energy of request nodes as ∞ for the absence of charging station at the request nodes ($g_i = \infty, \forall i \in \{R_1, \cdots, R_Z\}$); (ii) fixing the visiting time of each charging station node as the sum of the visiting time of request node and the travel time between charging station node and request node; (iii) setting the revenue of charging station nodes as 0 for the absence of request at the charging station nodes ($c_i = 0, \forall i \in \{C_1, \cdots, C_Z\}$), the modified graph can be easily accommodated by the proposed model.

B. Mathematical Formulation of EVRP

In this subsection, we elaborate the offline EVRP first. A fleet of EVs are coordinated to serve the transportation requests of customers within a given area. In this process, EVs operator has to make an optimal schedule with respect to the following four aspects of all EVs:

a) the total charging time, $\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} r_i^k g_i x_{ij}^k$ with x_{ij}^k being a binary variable for EV k choosing edge ij or not;
b) total charging cost, $\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} r_i^k p_i x_{ij}^k$;
c) travel time, $\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} T_{ij} x_{ij}^k$;
d) EV usage fee and revenue from serving transportation requests, $\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} c_i x_{ij}^k$ with c_i denoting a unified cost vector which represents both the EV usage fee c_{ev} and the negative serving cost $-\mathscr{M}_i$: $c_i = \begin{cases} -\mathscr{M}_i, & \text{if } i \in \mathcal{R} \\ c_{ev}, & \text{if } i = v_1. \end{cases}$

To represent decision maker's interests and preferences on time saving (travel time and charging time) and money saving (charging cost and EVs usage fee), the authors propose to use weighting method with a discount factor ω_T to characterize the attitude of EVs operator towards time-related terms and cost-related terms in objective function. Thus, the offline EVRP can be formulated into a mixed integer nonlinear program (MINLP) as below:

Problem 1 (MINLP)

$$\min_{x_{ij}^k\in\{0,1\}, r_i^k\in\mathbb{R}} \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} \Big\{\omega_T r_i^k g_i x_{ij}^k + r_i^k p_i x_{ij}^k + \omega_T T_{ij} x_{ij}^k + c_i x_{ij}^k\Big\} \tag{1}$$

$$\text{s.t.} \sum_{j\in\mathcal{V}} x_{ij}^k - \sum_{j\in\mathcal{V}} x_{ji}^k = b_i, \quad b_1 = 1, b_n = -1,$$
$$b_i = 0, \forall i \in \mathcal{V}\setminus\{v_1, v_n\}, k \in \mathcal{K} \tag{2}$$

$$\sum_{k\in\mathcal{K}}\sum_{j\in\mathcal{V}} x_{ij}^k \le 1, \quad \forall i \in \mathcal{R} \tag{3}$$

$$t_j \ge \left(T_{ij} + g_i r_i^k + t_i\right) x_{ij}^k, \quad \forall i \in \mathcal{V}\setminus v_n, \; j \in \mathcal{V}\setminus v_1, \; k \in \mathcal{K} \tag{4}$$

$$\tilde{E}_j^k = \sum_{i\in\mathcal{V}\setminus v_n} \left(E_i^k + r_i^k - e_{ij}\right) x_{ij}^k, \quad \forall j \in \mathcal{V}\setminus v_1, \; k \in \mathcal{K} \tag{5}$$

$$0 \le \tilde{E}_j^k \le E_{k,max}, \quad \forall j \in \mathcal{V}\setminus v_1, \; k \in \mathcal{K} \tag{6}$$

$$0 \le E_i^k \le E_{k,max}, \quad \forall i \in \mathcal{V}\setminus v_n, \; k \in \mathcal{K} \tag{7}$$

$$0 \le r_i^k \le r_{max}, \quad \forall i \in \mathcal{V}\setminus v_n, \; k \in \mathcal{K} \tag{8}$$

$$\tilde{E}_i^k = E_i^k \quad \forall i \in \mathcal{V}, \; k \in \mathcal{K}$$
$$E_{v_1}^k = E_0^k \quad \forall k \in \mathcal{K} \tag{9}$$

To meet the physical requirements of charging system and transportation system, network flow conservation constraints, battery energy constraints and pickup time constraints are defined as (2)-(9) in Problem 1. Constraint (2) indicates that all EVs are subject to the flow conservation constraint. In other words, an EV entering in the request node has to exit out at the same request node, and EV starts at the start depot while returning back to the end depot. Constraint (3) means each customer can be served at most once by the EV fleet. Note that some transportation requests may not be served by the fleet if the cost is higher than the benefit. Time constraints with prescribed pickup time t_j of customer and charging time $g_i r_i^k$ of EV are specified in (4), which states that the arrival time at request nodes should not be later than prescribed pickup time for transportation requests. Battery energy dynamics of EVs are characterized by equation (5) in which $e_{ij} = \epsilon d_{ij}$ denotes the amount of energy consumption on edge ij, and ϵ

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is the energy consumption rate and d_{ij} is the travel distance from node i to j, $i, j \in \mathcal{V}$. Constraints (6), (7) and (8) enforce lower and upper bounds on battery energy level of EV k, E_i^k and charged amount at node i, r_i^k. As for (6), to facilitate the construction of dual sub-problem of Problem1, $\tilde{E}_j^k$ is defined as a copy of variable E_i^k. The charging amount $r_{v_n}^k, \forall k \in \mathcal{K}$ is set as 0, since the charging at end depot after all transportation requests have been served is out of scope of the EVRP problem. The initial energy level of all EVs at the start depot is given in Eq.(9).

Lemma 1: The aforementioned Problem 1 is an NP-hard problem.

Proof: Given a notorious NP hard problem [44], the vehicle routing problem, we can reduce the vehicle routing problem to Problem 1 with the energy and time constraints omitted. Due to the NP hardness of vehicle routing problem, the NP hardness of Problem 1 is also proved. □

Due to the nonlinear and integer constraints, it requires massive computation resources and unbearable computation time to solve Problem 1 to optimality. To simplify Problem 1, in the next section, we first linearize nonlinear terms by the Big-M method, and then propose a decomposition based algorithm to handle the linearized version.

III. Solution of the Offline Problem—Benders Decomposition Framework

In practice, Problem 1 is a large-scale MINLP, thus obtaining its optimal solution is a highly time-consuming task. Therefore, solving it directly is not suitable for the online implementation. In this section, we pursue a distributed algorithm which achieves optimal solutions. To solve Problem 1 efficiently, we first linearize the nonlinear terms. Under the framework of Benders decomposition (BD), we then break down the resultant problem into two kinds of simpler problems, i.e., a master problem (MP) and a set of sub-problems (SP).

A. Linearization of MINLP Problem

Based on the Big-M method, by equivalently replacing constraints (4,5) with the following constraints (10) and (11), bilinear terms can be eliminated.

$$M(1-x_{ij}^k) \geq -\tilde{E}_j^k + E_i^k + r_i^k - e_{ij}x_{ij}^k \geq -M(1-x_{ij}^k), \quad \forall i \in \mathcal{V}\setminus v_n, j \in \mathcal{V}\setminus v_1, k \in \mathcal{K} \tag{10}$$

$$t_j \geq \left(T_{ij} + g_i r_i^k + t_i\right) - M(1-x_{ij}^k), \quad \forall i \in \mathcal{V}\setminus v_n, \; j \in \mathcal{V}\setminus v_1, k \in \mathcal{K} \tag{11}$$

Introducing nonnegative variables η_{ij}^k and ϵ_{ij}^k, bilinear terms in objective function can be eliminated by adding constraints (12).

$$\eta_{ij}^k \geq r_i^k g_i - M(1-x_{ij}^k), \quad \forall i \in \mathcal{V}\setminus v_n, \; j \in \mathcal{V}\setminus v_1, \; k \in \mathcal{K} \tag{12a}$$

$$\epsilon_{ij}^k \geq r_i^k p_i - M(1-x_{ij}^k), \quad \forall i \in \mathcal{V}\setminus v_n, \; j \in \mathcal{V}\setminus v_1, \; k \in \mathcal{K} \tag{12b}$$

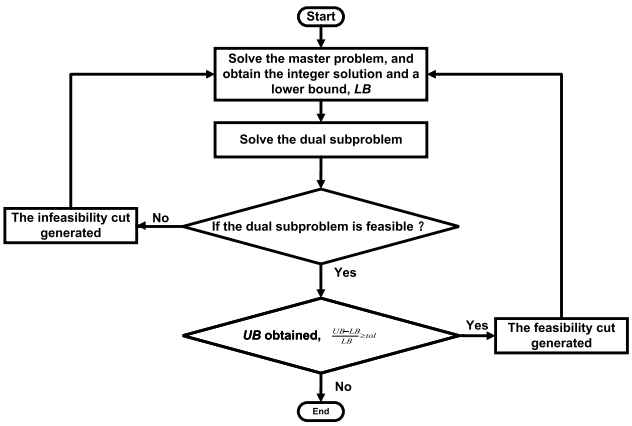


Fig. 3. Schematic flowchart of Benders decomposition.

Then, the original EVRP is equivalently converted into an MILP:

Problem 2 (MILP)

$$\min_{x_{ij}^k \in \{0,1\}, \eta_{ij}^k, E_{ij}^k, \epsilon_{ij}^k, r_i^k \in \mathbb{R}} \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} \left\{\omega_T \eta_{ij}^k + \epsilon_{ij}^k + \omega_T T_{ij} x_{ij}^k + c_i x_{ij}^k\right\}$$

s.t. (2)-(3) (6)-(12)

In general, Problem 2 is still time-consuming due to the NP-hardness of MILP. In order to solve Problem 2 in real time, a solution algorithm is devised in the next subsection to decompose Problem 2 into several simple problems. These problems include a set of SPs which are linear programs and can be solved distributedly, and an MP that can be relaxed and then solved by an equivalent linear program.

Remark 2: The scenario where there are multiple delivery requests from one physical node can also be accommodated by the proposed model. When there are multiple requests from one physical node, these requests can be split as separable virtual nodes, which connect to each other with zero travel distance.

B. Framework of Benders Decomposition Method

The basic idea behind BD shown in Fig. 3 is to decompose the complicating problem into two simpler problems, namely the MP and SP. The MP consists in finding optimal solution without technical validation of time constraints and battery energy constraints (4)-(9). Its solution yields a lower bound on the objective function of the original problem. The SP receives optimal solution of the MP and attempts to validate if it is technically feasible. Its solution yields an upper bound on the objective function of the problem and is used to generate Benders cuts for the MP. Then, the MP and SP are solved iteratively, until the upper and lower bound are sufficiently close [45].

1) Master Problem: In each iteration of BD, a Benders cut will be added to the constraint set of the MP. The MP with q infeasibility cuts and p feasibility cuts **(MP-2)** is formulated as follows.

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MP-2

$$\min_{x_{ij}^k \in \{0,1\}, Z} \quad Z$$
$$\text{s.t. } (2)-(3)$$
$$D^q(X) \le 0, \quad \forall q \in \mathcal{I} \tag{13a}$$
$$D^p(X) + \sum_{k\in\mathcal{K}} \sum_{i\in\mathcal{V}} \sum_{j\in\mathcal{V}} \left\{ \omega_T T_{ij} x_{ij}^k + c_i x_{ij}^k \right\} \le Z, \quad \forall p \in \mathcal{J}, \tag{13b}$$

where $\mathbb{X}_{mp}$ is the collection of decision variables x_{ij}^k, for simplicity $D^q(X)$ and $D^p(X)$ represent the infeasibility and feasibility Benders cuts respectively.

Remark 3: In MP-2, all EVs are assumed as homogeneous. To simplify the MP, next we drop the vehicle index k and present a new problem MP-3, which is equivalent to MP-2 as shown in Lemma 2.

MP-3

$$\min_{x_{ij}\in\mathbb{X}_{mp}, Z} \quad Z$$
$$\text{s.t. } \sum_{j\in\mathcal{V}} x_{ij} - \sum_{j\in\mathcal{V}} x_{ji} = b_i, \quad b_1 = K, b_n = -K,$$
$$b_i = 0, \forall i \in \mathcal{R} \tag{14}$$
$$\sum_{j\in\mathcal{V}} x_{ij} \le 1, \quad \forall i \in \mathcal{R} \tag{15}$$
$$\mu_1(q): D^q(X) \le 0, \quad \forall q \in \mathcal{I} \tag{16}$$
$$\mu_2(p): D^p(X) + \sum_{i\in\mathcal{V}} \sum_{j\in\mathcal{V}} \left\{ \omega_T T_{ij} x_{ij} + c_i x_{ij} \right\} \le Z, \quad \forall p \in \mathcal{J} \tag{17}$$

*Lemma 2: Under the assumption of homogeneous EVs, by combining the individual EV flow of **MP-2** as a bundled EV flow, **MP-2** can be converted as **MP-3** equivalently.*

Proof: In MP-2, with the homogeneous EVs, we can use decision variables associated with vertices, without the index of vehicle, to keep track of vehicle states, thus MP-2 can be reformulated as MP-3 equivalently. □

2) Sub-Problems and Benders Cuts: Given the integer MP solution, SP can be formulated as follows

$$\min_{\eta_{ij}^k, E_i^k, \epsilon_{ij}^k, r_i^k \in \mathbb{R}} \sum_{k\in\mathcal{K}} \sum_{i\in\mathcal{V}} \sum_{j\in\mathcal{V}} \left\{ \omega_T \eta_{ij}^k + \epsilon_{ij}^k \right\}$$
$$s.t. \quad (6)\text{-}(12)$$

Let $\mathbb{X}_{sp}^k$ denote all decision variables of **SP-k**. Note that SP can be implemented in a distributed manner, i.e., each vehicle k solves the following linear program independently.

SP-k

$$\min_{\mathbb{X}_{sp}^k} \sum_{i\in\mathcal{V}} \sum_{j\in\mathcal{V}} \left(\omega_T \eta_{ij} + \epsilon_{ij} \right)$$
$$s.t. \ (6)-(12) \quad \text{on each vehicle k.}$$

Then dual variables $(\lambda_{ijk}^1, \cdots, \lambda_{ijk}^5, \lambda_{ik}^6, \cdots, \lambda_{ik}^9)$ are introduced for each SP-k, where $(\lambda_{ijk}^1, \lambda_{ijk}^2)$ correspond to the constraint (6), $(\lambda_{ijk}^3, \lambda_{ijk}^4, \lambda_{ijk}^5)$ correspond to (7)-(9), $(\lambda_{ik}^6, \cdots, \lambda_{ik}^9)$ correspond to constraints (10)-(12). To produce the Benders cut, the dual problem of SP-k is derived as follows.

Dual SP-k

$$\max_{\Lambda_k} \quad \Phi^{UB}(\Lambda_k) \quad s.t. \text{ constraints of dual SP-k}$$

where $\Lambda_k = \{\lambda_{ijk}^1, \cdots, \lambda_{ijk}^5, \lambda_{ik}^6, \cdots, \lambda_{ik}^9\}$, and $\Phi^{UB}(\Lambda_k)$ is shown in equation (18), shown at the bottom of the page.

Given the solution Λ_k of Dual SP-k, Benders cuts $D^w(X), \forall w \in \{q, p\}$ in (13) can be derived,[2] where $D^w(X) = f^w(X) + c^w$:

$$f^w(X) = \sum_{i\in\mathcal{V}\setminus v_n} \sum_{j\in\mathcal{V}\setminus v_1} \Big\{ (\lambda_{ij}^1 + \lambda_{ij}^4 + \lambda_{ij}^5) M + \lambda_{ij}^2 (M + e_{ij}) + \lambda_{ij}^3 (M - e_{ij}) \Big\} x_{ij}$$

$$c^w = \sum_{i\in\mathcal{V}\setminus v_n} \sum_{j\in\mathcal{V}\setminus v_1} \Big\{ \lambda_{ij}^1 \big\{ \big(T_{ij} + t_i\big) - M - t_j \big\} - M\lambda_{ij}^2 - M\lambda_{ij}^3 - M\lambda_{ij}^4 - M\lambda_{ij}^5 \Big\} - \sum_{j\in\mathcal{V}\setminus v_1} \lambda_j^6 E_{max} - \sum_{j\in\mathcal{V}\setminus v_1} \lambda_i^7 E_{max} - \sum_{i\in\mathcal{V}\setminus v_n} \lambda_i^8 r_{max} - \sum_{i=v_1} E_0 \lambda_i^9$$

IV. Solution of the Offline Problem—Alternative Formulation of MP

Note that a major computational bottleneck still exists, as the MP-3 is also a mixed integer linear program (MILP). By observation, there is an important conclusion that constraint matrix of the initial MP-3 satisfies the property of total unimodularity (TU) [46]. Therefore, in order to speed up the proposed algorithm, a Lagrangian relaxation method is applied to solve the MP-3 by relaxing generated cuts. A near-optimal solution is obtained. In addition, a novel time-related valid cut is further added into the MP for a faster convergence.

[2]The details of a simple example of Benders decomposition are shown in Appendix A.

$$\Phi^{UB}(\Lambda_k) = \sum_{i\in\mathcal{V}\setminus v_n} \sum_{j\in\mathcal{V}\setminus v_1} \Big\{ \lambda_{ij}^1 \big\{ \big(T_{ij} + t_i\big) - M(1 - x_{ij}) - t_j \big\} + \lambda_{ij}^2 \big\{ e_{ij} x_{ij} - M(1 - x_{ij}) \big\} + \lambda_{ij}^3 \big\{ - e_{ij} x_{ij} - M(1 - x_{ij}) \big\} + \lambda_{ij}^4 \big\{ - M(1 - x_{ij}) \big\} + \lambda_{ij}^5 \big\{ - M(1 - x_{ij}) \big\} \Big\} - \sum_{j\in\mathcal{V}\setminus v_1} \lambda_j^6 E_{max} - \sum_{j\in\mathcal{V}\setminus v_1} \lambda_i^7 E_{max} - \sum_{i\in\mathcal{V}\setminus v_n} \lambda_i^8 r_{max} - \sum_{i=v_1} E_0 \lambda_i^9 \tag{18}$$

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A. Acceleration of MP Computation

Despite the decomposition into simpler problems, the master problem MP-3 still remains an MILP. However, when there is no cuts (16),(17) added, the feasible region of MP-3 satisfies the property of TU which can be helpful in designing computational efficient solution algorithm [46]. Hence, by exploiting the property of TU, we propose acceleration schemes for MP-3 in this subsection.

1) Lagrangian Relaxation of MP-3: Instead of solving MP-3 exactly, we approximately solve a relaxed problem MP-4 as follows:

MP-4

$$\min_{x_{ij}\in\{0,1\},Z} \quad \mathcal{L}_{MP(x_{ij},Z,\mu_1,\mu_2)}$$
$$\text{s.t. } (14), (15)$$

where

$$\mathcal{L}_{MP(x_{ij},Z,\mu_1,\mu_2)} = Z + \sum_{q\in\mathcal{I}} \mu_1(q) D^q(X) + \sum_{p\in\mathcal{J}} \mu_2(j)\Big\{D^p(X) + \sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}}\Big\{\omega_T T_{ij} x_{ij} + c_i x_{ij}\Big\} - Z\Big\} \quad (18)$$

μ_1 and μ_2 are Lagrange multipliers corresponding to (16) and (17) respectively.

The definition of total unimodular at subsection *19.1 Total unimodularity and optimization* in book [46] is introduced firstly.

Definition 1: A matrix A is totally unimodular [46] if each subdeterminant of A is $0, +1,$ *or* -1. *In particular, each entry in a totally unimodular matrix is* $0, +1,$ *or* -1.

Note that solving MP-4 is equivalent to solving a linear program (LP), which greatly reduces the computation burden. This claim is proved in the following lemma:

Lemma 3: MP-4 can be exactly relaxed to an LP, in the sense that their solutions are equivalent.

Proof: The relaxed linear program of integer program would have only integer solutions, which happens when the constraint matrix is totally unimodular [47].

When Benders cut (16) and (17) are relaxed, the constraint matrix of the resultant problem MP-4 is the standard form of the single depot vehicle scheduling problem for which its constraint matrix satisfied the property of total unimodularity [48]. Then, $P = \{x|(14), (15)\}$ satisfies the property of TU as well, MP-4 can be exactly relaxed as a linear programming problem without loss of optimality. □

2) Valid Cut for Master Problem: Although the original complicating MILP problem is decomposed into the simpler MP and SP, the time complexity of BD method is still non-polynomial because of the binary variables in MP. In order to accelerate the convergence of the BD process, a novel kind of valid cuts (19) has been added to the constraint set of MP-4:

$$t_j \geq T_{ij} + t_i - M(1 - x_{ij}), \quad \forall i \in \mathcal{V}\setminus v_n, \ \ j \in \mathcal{V}\setminus v_1 \quad (19)$$

The basic idea is that valid cut (19) lops off part of the feasible region in MP-4, but no optimal solution of the original Problem 2 is excluded. Therefore, it reduces the number of iterations without loss of optimality. This is rigorously proved in the following lemma.

Lemma 4: The addition of valid cut (19) neither breaks the TU property of MP-4, nor excludes any optimal solution of Problem 2.

Proof: Note that $t_i, i \in \mathcal{R}$ is a parameter given by customers rather than a decision variable. It is easy to see valid cut (19) is equivalent to

$$0 \geq \big(T_{ij} - \big(t_j - t_i\big)\big)\, x_{ij}, \quad (20)$$

then we have:

- If $\big(T_{ij} - \big(t_j - t_i\big)\big) > 0$, (20) is equivalent to $x_{ij} = 0$, adding (20) does not break the TU property of MP-4. Moreover, in this case, from (11), the optimal solution x_{ij}^k of Problem 2 must equal to 0, which satisfies (20).
- If $\big(T_{ij} - \big(t_j - t_i\big)\big) \leq 0$, (20) will be satisfied no matter what value x_{ij} is, thus it is a redundant constraint.

Hence, any optimal solution of Problem 2 will certainly satisfy (20), and the TU property is not violated. □

B. Proposed Offline EVRP Algorithm

In this subsection, a computation-efficient algorithm is proposed to optimize the MP-4 firstly, then the framework of modified Benders decomposition (MBD) method is characterized in detail.

1) Dual Lagrangian Multiplier Algorithm (DLMA) for MP-4: To solve MP-4 efficiently, a Lagrangian multiplier coordination based algorithm is proposed. Denote $\mathbb{X}_{mp4} = \big\{x_{ij}\,|x_{ij} \in [0,1], (14), (15), (20)\big\}$ as the feasible region of MP-4. Every iteration, the dual variables $\mu = \{\mu_1, \mu_2\}$ are updated according to the following rule: $\mu = \max\{\mu + \Delta\mu, 0\}$,

$$\Delta\mu_1(q) = \delta_1 \left[D^q(\bar{X})\right]^+_{\mu_1(q)}, \quad \forall q \in \mathcal{I} \quad (21)$$

$$\Delta\mu_2(p) = \delta_2 \left[D^p(\bar{X}) + \sum_{i,j}\big\{\omega_T T_{ij}\bar{x}_{ij} + c_i\bar{x}_{ij}\big\} - \bar{Z}\right]^+_{\mu_2(p)}, \quad \forall p \in \mathcal{J}, \quad (22)$$

where δ_1 and δ_2 are non-negative step sizes, and the function $[f]_b^+$ is defined as

$$[f]_b^+ = \begin{cases} \max\{f, 0\}, & \text{if } b \leq 0 \\ f, & \text{otherwise} \end{cases}$$

$\bar{x} = arg\min_{x\in\mathbb{X}_{mp4}} \mathcal{L}_{MP}$, and $\bar{Z}$ is given by

$$\bar{Z} = \max_{p\in\mathcal{J}} \Big\{D^p(X) + \sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}}\Big\{\omega_T T_{ij} x_{ij} + c_i x_{ij}\Big\}\Big\}. \quad (23)$$

Detailed process of *DLMA* is shown in Algorithm 1 in which $\zeta = (\bar{x}, \bar{Z})$ is defined as a solution of MP-4 and $c(\zeta)$ is the objective value of MP-4.

Remark 4: In Algorithm 1, μ_{max} *is set to be sufficiently large to ensure the optimal solution of MP-4 is always feasible for MP-3.*

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Algorithm 1 DLMA for MP-4

Initialization: $ite = 1, \mu = \mu_{max}$,
while $\exists i \in \mathcal{I}, j \in \mathcal{J} : |\Delta\mu_1(i)|, |\Delta\mu_2(j)| > tol$ and $ite < ITE$ **do**
  Set $\bar{x} = arg\min_{x \in \mathbb{X}_{mp4}} \mathcal{L}_{MP}$.
  Obtain $\bar{Z}$ according to (23)
  Solution $\zeta_h = \{\bar{x}, \bar{Z}\}$ is obtained, $ite = ite + 1$
  Set $\Delta\mu$ according to (21),(22) and $\mu = \max\{\mu + \Delta\mu, 0\}$
  if $\Delta\mu_1(i) = 0, \forall i \in \mathcal{I}$ and $\Delta\mu_2(j) = 0, \forall j \in \mathcal{J}$ **then**
    return $\zeta = \zeta_h$
  else if ζ_h is feasible and $c(\zeta_h) < c(\zeta)$ **then**
    $\zeta = \zeta_h$
  else if ζ_h is not feasible **then**
    $\mu = \mu_{max}$
  end if
  $ite = ite + 1$
end while

Algorithm 2 MBD Algorithm

Initialization : i = 1, tol = ϵ, $\psi^{IC} = \emptyset$, $\psi^{FC} = \emptyset$,
Obtain initial $x \in \mathbb{X}_{mp}$
while 1 **do**
  Solve the dual SP-k for each EV k in a distributed manner, and obtain Lagrangian multipliers, $\Lambda(i) = \sum_{k \in \mathcal{K}} \Lambda_k(i)$, and ***upper bound*** of Problem 2.
  if The dual subprobem has unbounded solution **then**
    $\psi^{IC} = \psi^{IC} \cup \{\Lambda(i)\}$; Construct Benders cuts (16).
  else if The dual subprobem has Feasible solution **then**
    if $\frac{UB-LB}{LB} \leq tol$ **then**
      Optimal solution obtained
      break
    else
      $\psi^{FC} = \psi^{FC} \cup \{\Lambda(i)\}$; Construct Benders cuts (17)
    end if
  end if
  Solve the master problem by *Algorithm 1*, and obtain the ***lower bound*** of Problem 2.
  $i = i + 1$
end while

2) MBD Algorithm: Based on the proposed decomposition formulation, an iterative distributed algorithm is developed in this subsection to solve the offline EVRP Problem 2. Algorithm 2 shows the details of MBD based method. Two sets containing infeasibility cuts and feasibility cuts are denoted by ψ^{IC} and ψ^{FC} respectively.

Remark 5: The proposed MBD algorithm enables the distributed implementation of the dual subproblem, as illustrated in Fig. 4. Due to the superiority of MBD in iteratively optimizing LPs, the computation speed of MBD is much faster than the dual decomposition approach proposed in the literature [15].

V. Online Routing Strategy

In order to handle uncertainties arising from future transportation requests, electricity price and travel time, the proposed MBD algorithm is embedded within the framework of rolling horizon to solve the online EVRP. In rolling horizon framework, system parameters should be updated every time step to capture the real-time change. Then, we shall re-optimize the problem after the revelation of new information.

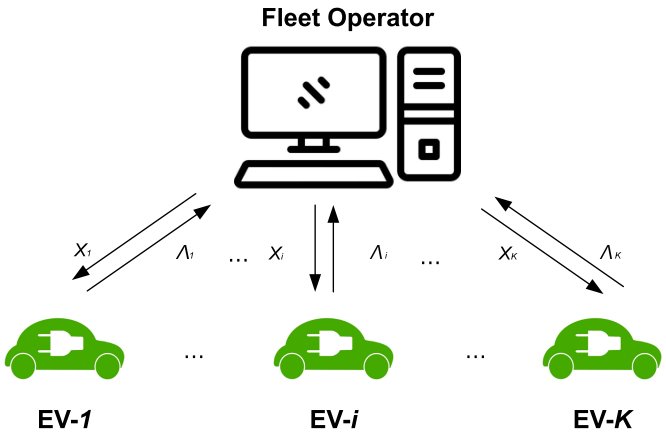


Fig. 4. Distributed implementation of MBD.

In every re-optimization stage, the start location of each EV will change, rendering the offline EVRP to be a multiple start depot problem. However, our proposed MBD algorithm is designed for the single depot problem, thus posing a new challenge on the practical implementation. To this end, we prove that under an equivalent transformation by introducing virtual depot, our proposed MBD algorithm can be directly adapted to the multi-depot case. The detailed transformation process is shown in Fig. 5.

Transformation Rule. *A virtual depot is introduced and connected with current location of all EVs. The travel time and energy consumption between the virtual depot and current location are set as zero and the energy difference between battery capacity and current energy level, respectively.*

Theorem 1: Following the transformation rule, multiple start depots routing problem can be equivalently transformed into the single depot case.

Proof: Define a set $\Psi^\tau = \{L_1^\tau, \cdots, L_K^\tau\}$ consisting of the start location of EVs at the τth time slot and a set $\Phi^\tau = \{R_1^\tau, \cdots, R_K^\tau\}$ consisting of the next request nodes to serve of all EVs at the τth time slot. Additional constraints (24a) have to be added to MP-4 to force that every road between the virtual depot and current start location must be travelled. Similarly, (24b) ensures that every road between the current start location and the next request node must be travelled.

$$x_{1i} = 1, \quad \forall i \in \Psi^\tau \tag{24a}$$

$$x_{L_i^\tau R_i^\tau} = 1, \quad \forall i \in \mathcal{K} \tag{24b}$$

Constraint (24) is equivalent to the following formulas:

$$\begin{cases} 1 \leq x_{ij} \leq 1, & \text{if } i = v_1, \ j \in \Psi^\tau \\ 1 \leq x_{L_i^\tau R_i^\tau} \leq 1, & \forall i \in \mathcal{K} \\ x_{ij} \in \{0, 1\}, & \text{else} \end{cases} \tag{25}$$

Clearly, (25) does not affect the TU property of MP-4, thus can be easily incorporated into our proposed framework. □

Now we present the details of online modified Benders decomposition (OMBD) method with virtual depot under the rolling horizon framework in Algorithm 3.

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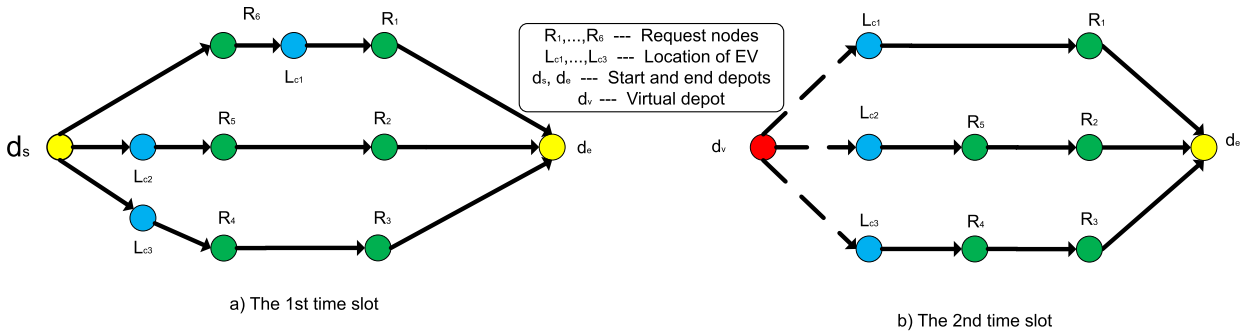


Fig. 5. Multi-depot Transformation; solid line (physical road) and dashed line (virtual road); the red circle (virtual depot), the green circle (request node), the yellow circle (physical depot), and the blue circle (end location of EVs at 1st time slot, i.e., the start location at 2nd time slot).

Algorithm 3 OILP Algorithm

Initialization: $\tau = 1$, overall time slots (T), initial SOC,
while $\tau \leq T$ **do**
  Broadcast the information of electricity charging price, charging capability and transportation requests of the τth time slot.
  if $\tau = 1$ **then**
    Solve offline EVRP by *Algorithm 2*.
  else if $\tau \neq 1$ **then**
    A virtual depot is introduced according to *Transformation rule*.
    Solve offline EVRP by *Algorithm 2*.
  end if
  $\tau = \tau + 1$
end while

VI. Performance Evaluation

In this section, the performance of proposed algorithm is evaluated. The parameter settings are introduced firstly, followed by the numerical results on the offline and online EVRP.

A. Parameter Settings

We first adopt the map data of Belgium,[3] which records the geographic coordinates of 1000 nodes, and construct its transportation network. The battery capacity and charging duration per unit energy are selected as 90 $kW \cdot h$, 1/150(fast charging)/ 1/22(slow charging) $hr/kW \cdot h$ (Tesla Model X [4]). In addition, the energy consumption rate and average speed of EVs are set as 0.24 $kW \cdot h/km$ and 60 km/h respectively. According to the electricity price of Belgium,[5] there are two kinds of charging prices \$0.299/$kW \cdot h$ and \$0.129/$kW \cdot h$, respectively. Due to the difference in charging capability, fast charging power comes with higher charging price and vice versa.[6] The usage cost of EV represents the monetary loss during the usage process of EV, such as wear and tear, depreciation, and its value is set as \$199. To characterize a practical scenario, we sample the positive revenue from a truncated normal distribution with a mean of \$9.05 and a standard deviation of \$5.[7] The following numerical experiments are all implemented 10 times, and statistical results are shown. For example, when the size of network is 20, these 20 nodes will be randomly selected from the 1000-node map at each run of simulation experiments. In this case, due to the different road topology of randomly selected nodes, the performance of online MBD on different road topology is evaluated. All optimization methods are implemented with MATLAB R2018b and Python 3.7 on Dell PowerEdge T640 2.3GHz CPU with 128 GB of memory.

[3]http://www.vrp-rep.org/datasets/item/2017-0001.html
[4]https://en.wikipedia.org/wiki/Tesla_Model_X
[5]https://www.globalpetrolprices.com/Belgium/electricity_prices/
[6]https://www.currentautomotive.com/a-guide-to-teslas-supercharging-network/

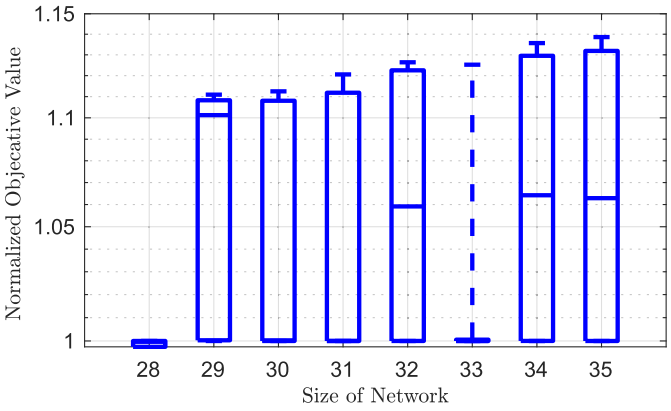


Fig. 6. Comparison on optimality.

B. Numerical Results of Offline EVRP

With the increasing size of network (from 28 to 35),[8] the numerical simulations have been performed 10 times for MBD, MBD without cut and MILP.

1) Optimality Gap Comparison: Compared with MILP based approach which can obtain global optimal solution for offline EVRP, MBD achieves near-optimal solution, as shown in Fig. 6. Note that the largest optimality gap is less than 15% over all instances. In some cases, even zero optimaility gap can be obtained with MBD.

2) Computation Time Comparison: The absolute computation time is illustrated in Fig. 7. It is obvious that our proposed MBD algorithm has faster computation speed. This performance gain becomes more prominent with the increase of

[7]https://www.fedex.com/en-us/shipping/one-rate.html
[8]There are only 2 EVs considered.

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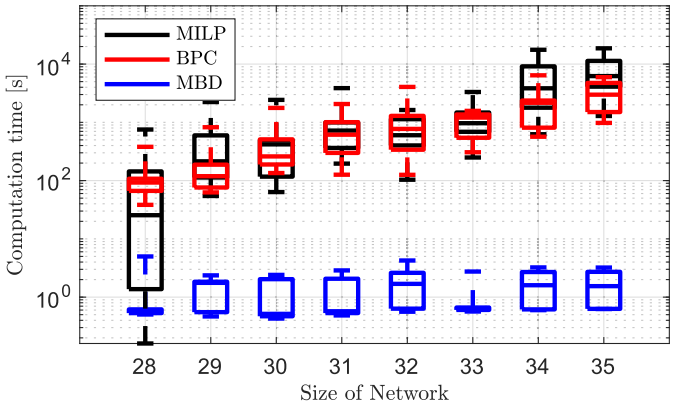


Fig. 7. Comparison on computation time.

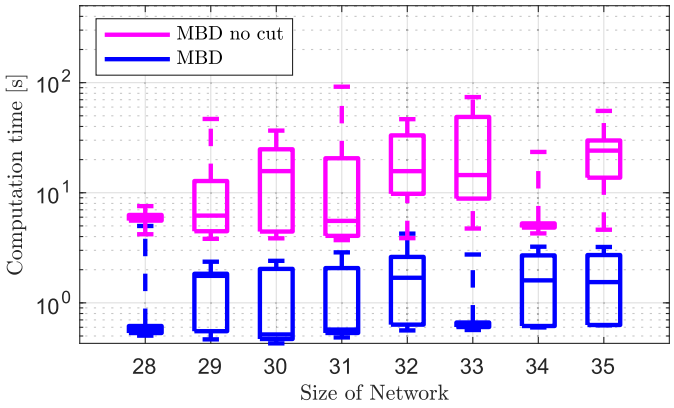


Fig. 8. Performance of proposed valid cut (19).

network size. Therefore, the proposed MBD is suitable for practical implementation of large scale problems. In addition, to validate the superiority of the proposed algorithm over the state-of-the-art algorithm, a comparison between the proposed algorithm and the branch price and cut (BPC, colored in red in Fig. 7.) method which is commonly used in numerous related works [6], [28], [29], [49]–[51], is conducted and illustrated in Fig. 7.

To validate the acceleration effect of valid cut (19), numerical simulations a) with and b) without the proposed valid cut, are also performed and the simulation results are shown in Fig. 8. We can conclude from the figure that the introduction of valid cut indeed improves the computation speed of MBD algorithm.

3) Scalability of Proposed MBD: To investigate the scalability of MBD, we extend the problem size to 350 customers and 35 EVs. The simulations have also been performed 10 times. As shown in Fig. 9, MBD can solve the problem in the order of seconds.

C. Numerical Results of Online EVRP

In simulation part, the entire rolling horizon consists of 5 time slots with each slot set as 0.5 hour. We assume that there are 8 requests occurred at the first time slot, and every time slot 2 new requests emerge. Here, the simulated competitive ratio is defined as the ratio between the objective value of proposed online MBD algorithm and the objective value of offline MILP, which showcases the performance of OMBD

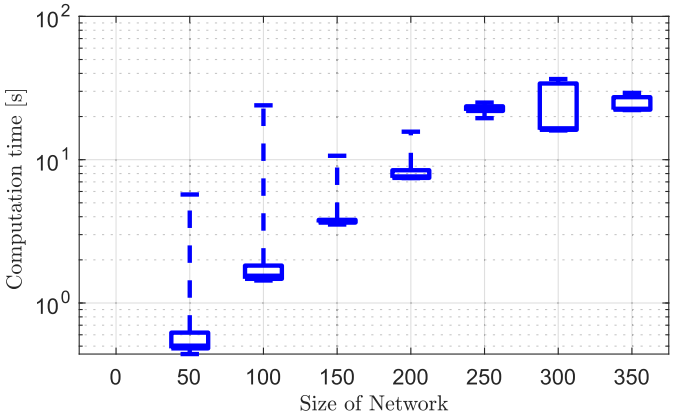


Fig. 9. Scalability of MBD.

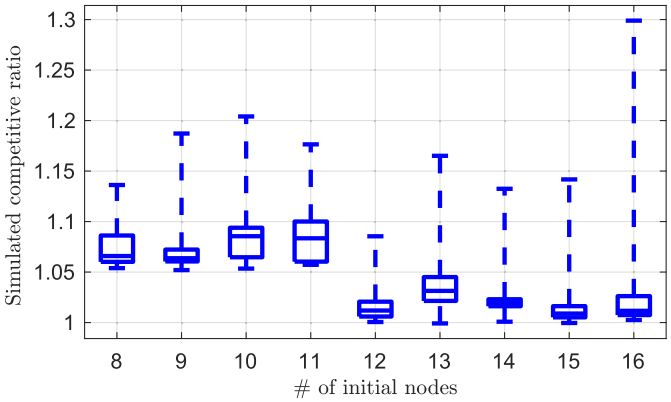


Fig. 10. Simulated competitive ratio over different road topology.

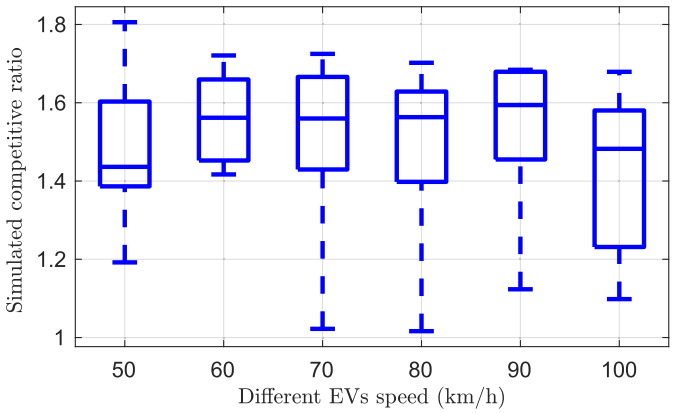


Fig. 11. Simulated competitive ratio over different driving speed.

method. The network size used in this subsection is limited in small size due to the difficulty of applying the offline MILP method over the whole time horizon.

1) Competitive Analysis and Robustness of OMBD: In the online settings, various scenarios with different number of initial requests (from 8 to 17) are considered. The OMBD and offline MILP algorithm where the information of all nodes occurred in the entire rolling horizon is disclosed at the first time slot, are implemented respectively. Since nodes are randomly chosen from a 1000-node map, each run of simulation has its own road topology. As shown in Fig. 10, compared with Off-MILP, the proposed online algorithm achieves a good

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performance with the simulated competitive ratio less than 1.35 over different road topology.

To evaluate the robustness of OMBD on different driving speed (the number of nodes is set as 8 at the first time slot, and every time slot 2 new requests emerge), different scenarios with changing EVs speed (from 50km/h to 100 km/h) are tested and the results are illustrated in Fig. 11. The largest simulated competitive ratio is still lower than 1.85 which shows that OMBD can achieve good performance on different driving speed. Besides, the simulation results in Fig. 11 showcase that the proposed model can accommodate various road conditions.

VII. Conclusion

In this paper, the rolling-horizon framework is used to handle the online EVRP, where a corresponding offline EVRP is solved repeatedly with the known customer requests and current traffic information, and only the first time step actions are implemented. In order to solve this offline EVRP efficiently in real-time, a BD based algorithm is proposed to decompose the original MINLP into simpler problems. As the resultant MP is still an MILP, we approximately solves it by the Lagrangian relaxation method, and validates its optimality gap in the simulation results. A novel kind of valid cut is also added to the MP so that the number of iterations is further reduced. This proposed MBD method is proved to be easily adapted to the online setting. Simulation results demonstrate that our proposed decomposition method outperforms the MILP based approach in terms of solution speed, while the optimality loss is minor.

Appendix A
A Simple Example of Benders Decomposition Method

Considering a MIP problem

$$\begin{aligned} \min_{x,y}\ & c^T x + f^T y \\ s.t.\ & Ax + By \geq b \\ & y \in \mathbb{B} = \{0,1\}^n \\ & x \geq 0 \end{aligned}$$

a subproblem can be obtained when the value of binary variable y is fixed as $\bar{y}$.

$$\begin{aligned} \min_x\ & c^T x \\ s.t.\ & Ax \geq b - B\bar{y} : u \\ & x \geq 0 \end{aligned}$$

With the dual variable u, the dual subproblem is:

$$\begin{aligned} \max_u\ & (b - B\bar{y})u \\ s.t.\ & A^T u \geq c \\ & u \geq 0 \end{aligned} \qquad \text{(DSP)}$$

Then, the master problem can be stated as follows

$$\begin{aligned} \min_y\ & f^T y + \phi \\ s.t.\ & y \in \mathbb{B} \\ & (b - By)^T \bar{u} \leq 0 : (\textit{Unbounded DSP}) \\ & (b - By)^T \bar{u} \leq \phi : (\textit{Feasible DSP}) \end{aligned}$$

where $\bar{u}$ is the solution of DSP.

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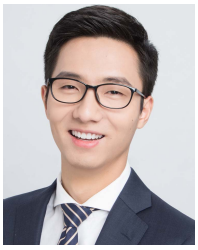
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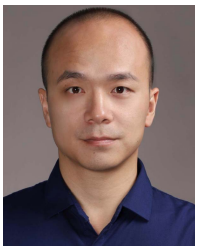
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